

Emotion Detection & Learning Support Engine

Submitted By

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PROJECT OVERVIEW

Purpose

The Emotion Detection Learning Support Engine is an AI-powered application designed to identify students' emotional states from textual input and provide personalised learning guidance. The system leverages deep learning models (BiLSTM and BERT) for emotion classification and uses the Gemini API to generate supportive educational responses.

Features

- Emotion detection from text input
- BiLSTM-based emotion classification
- BERT-based emotion classification
- Personalised learning recommendations
- Gemini AI-generated guidance
- Offline fallback responses
- Analytics dashboard
- CSV interaction logging
- Interactive Streamlit interface

ARCHITECTURE

Frontend

The frontend is developed using **Streamlit**, providing a simple and interactive interface where users can:

- Enter text describing their emotions
- Submit input for analysis
- View detected emotion
- View confidence score

- Receive personalised guidance
- View analytics dashboard

Backend

The backend is implemented in **Python**.

Major components include:

- Text preprocessing
- BiLSTM model
- BERT model
- Emotion prediction module
- Gemini API integration
- Logging module
- Analytics generation

Database

The project currently uses **CSV files** for storing interaction logs.

Stored information includes:

- User input
- Predicted emotion
- Confidence score
- Generated response
- Timestamp

Future versions may integrate MongoDB or PostgreSQL.

SETUP INSTRUCTIONS

Prerequisites

Install:

- Python 3.10+
- pip
- Virtual Environment
- Git

Install Python libraries

- Streamlit
- Tensorflow
- Torch
- Transformers
- scikit-learn
- pandas
- numpy
- matplotlib
- python-dotenv
- google-generativeAI

Installation

Clone repository

- git clone <https://github.com/Sushma-1510/Emotion-detection-learning-support-engine.git>

Go inside project

- `cd Emotion-detection-learning-support-engine`

Create virtual environment

- `python -m venv venv`

Activate environment

Windows

- `venv\Scripts\activate`

Linux/Mac

- `source venv/bin/activate`

Install dependencies

- `pip install -r requirements.txt`

Configure environment

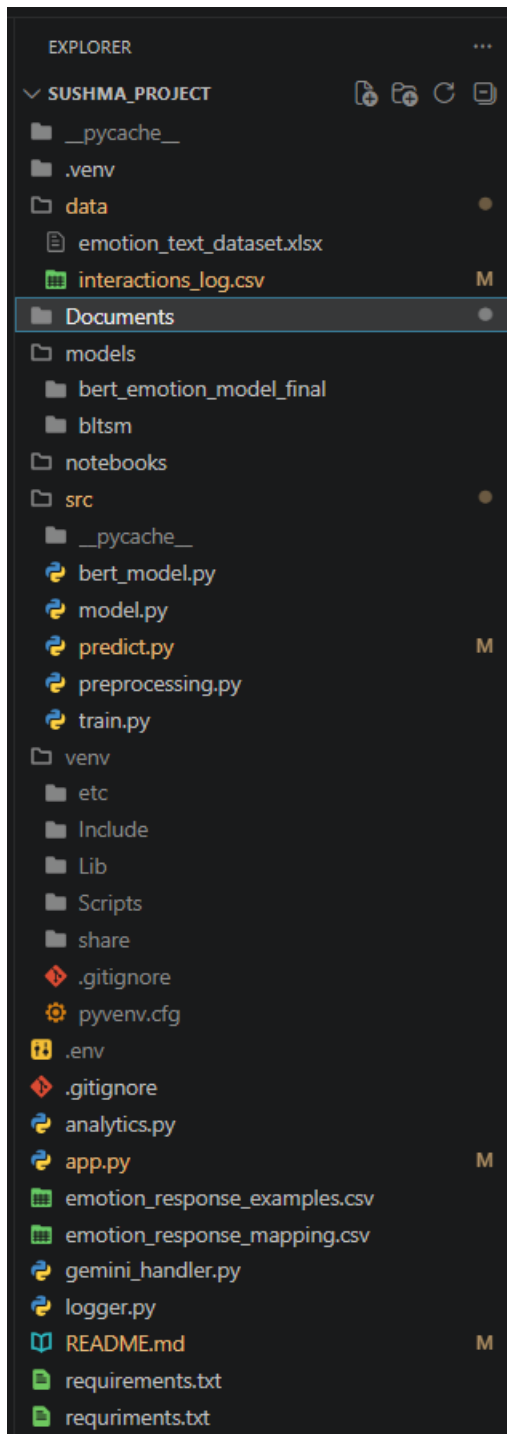
Create

- .env

Add

- GOOGLE_API_KEY=YOUR_API_KEY

FOLDER STRUCTURE



RUNNING THE APPLICATION

Activate virtual environment

- `venv\Scripts\activate`

Run application

- `streamlit run app.py`

Application URL

- `http://localhost:8501`

API DOCUMENTATION

Although the application primarily runs locally, the following APIs/modules are used.

Module	Function
BiLSTM Predictor	Predicts emotion
BERT Predictor	Predicts emotion
Gemini API	Generates personalized responses
Logger	Stores interaction history

Example Input

```
{  
  "text": "I feel stressed about my exams."  
}
```

Example Output

```
{  
  "emotion": "Anxiety",  
  "confidence": "94%",  
  "response": "Take short study sessions with regular breaks..."  
}
```

AUTHENTICATION

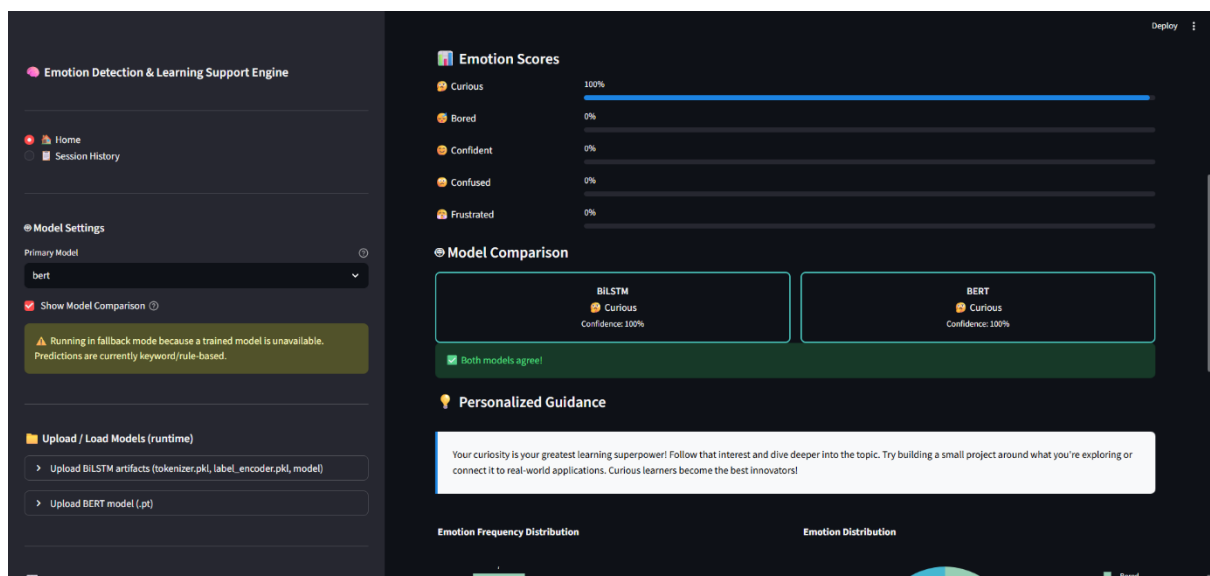
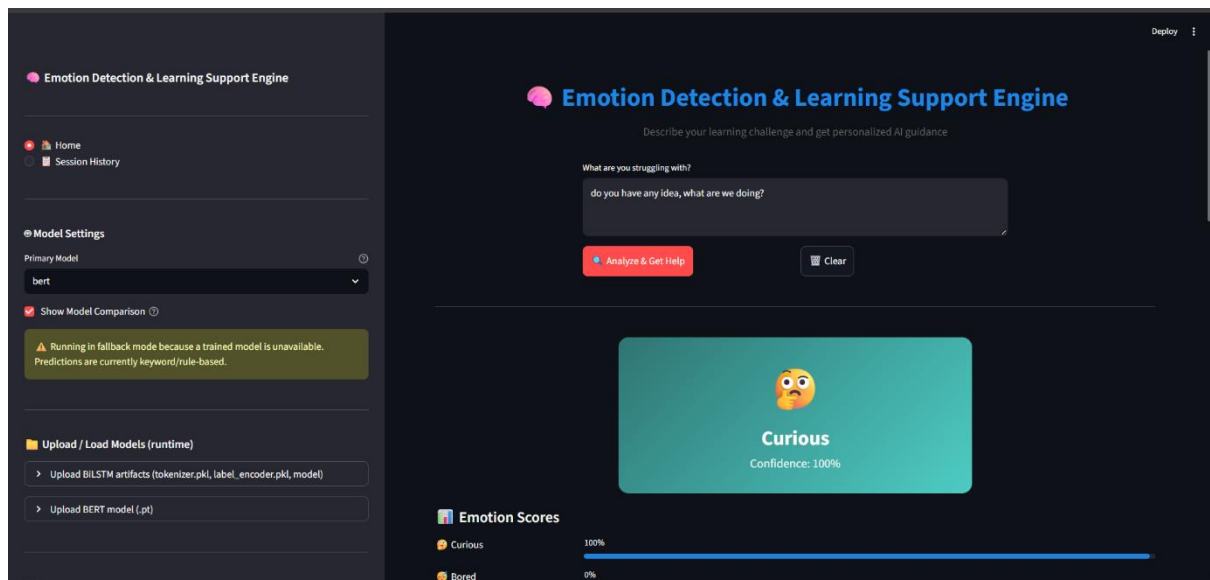
Current Version

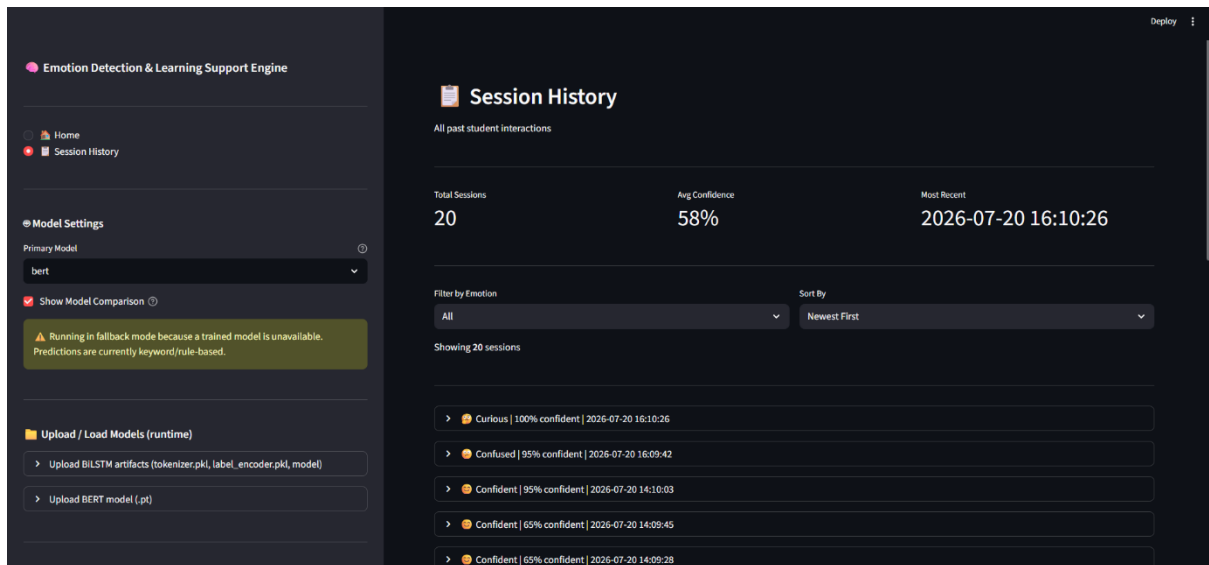
- No user authentication
- Local execution only

Future Version

- User Login
- JWT Authentication
- Role-based access
- Secure cloud deployment

RESULTS





TESTING

Testing performed

- Unit Testing
- Model Prediction Testing
- Functional Testing
- User Acceptance Testing (UAT)
- Integration Testing

Tools

- Streamlit
- Python
- TensorFlow
- PyTorch

FUTURE ENHANCEMENTS

- Voice emotion detection
- Facial emotion recognition
- Multilingual support
- Student progress tracking
- Teacher dashboard
- Mobile application
- Cloud deployment
- MongoDB integration
- User authentication
- Real-time analytics
- Personalised learning history
- Explainable AI for emotion prediction

REFERENCE

GitHub Repository

<https://github.com/Sushma-1510/Emotion-detection-learning-support-engine>

Code Demonstration:

https://drive.google.com/file/d/12LUMbhw3RC3hM4wf81Nxz40rjGJp4m64/view?usp=drive_link

Project Demonstration:

https://drive.google.com/file/d/1vUuiVp43V1kdr24NDTDCOvwIAvwGeF4E/view?usp=drive_link